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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

EXPERIMENT – 1

Title: IAM User with Programmatic Access and AWS Free Tier Billing Alert

AIM

To create an AWS IAM user with programmatic access only, generate and securely download an access key pair, and configure billing/Free Tier notifications so that an email alert is sent when usage or spending approaches the defined limit.

DESCRIPTION

AWS Identity and Access Management (IAM) is used to create users and control their access to AWS services. Programmatic access uses an access key ID and secret access key for tools or applications; it does not require an IAM console password. AWS Budgets and Free Tier alerts help monitor usage and costs. Access keys are sensitive credentials and must never be shared or included in a lab report.

REQUIREMENTS

1. An AWS account with permission to manage IAM users and billing settings.
2. A valid email address for billing notifications.
3. A web browser and internet connection.
4. Secure storage for the downloaded access-key CSV file.

PROCEDURE / STEPS

1. Sign in to the AWS Management Console using an account authorized to manage IAM.

2. Open IAM → Users → Create user. Enter the required IAM username (the supplied screenshots show the example username “test”).

3. On the permissions step, assign only the permissions required for the lab. Follow the principle of least privilege; do not grant broad administrator access unless explicitly authorized by the instructor.

4. Complete user creation and open the newly created IAM user. Confirm that console access is disabled, so the user is intended for programmatic access only.

5. Open the Security credentials tab and select Create access key.

6. Choose the Command Line Interface (CLI) use case (or the use case specified by the lab), acknowledge the recommendation, and create the key.

7. On the Retrieve access keys page, download the CSV file. AWS displays the secret access key only at creation time. Store the CSV securely and do not upload it to the lab record, email it, or commit it to source control.

8. Open Billing and Cost Management → Budgets → Create budget. Select the Zero spend budget template if the objective is to receive a notification when account spending first exceeds the zero-spend threshold.

9. Complete the budget setup and specify the email recipient(s) for notifications. Review the notification threshold and save the budget.

10. Separately open Billing → Free Tier / Billing preferences and enable Free Tier usage alerts, entering or confirming the notification email address. Verify that the email subscription/notification setting is active.

11. Check the Budgets page to confirm that the budget appears in the list. Check billing preferences and the email inbox to verify the Free Tier alert subscription and any confirmation message.

SCREENSHOT EVIDENCE – IAM USER CREATION

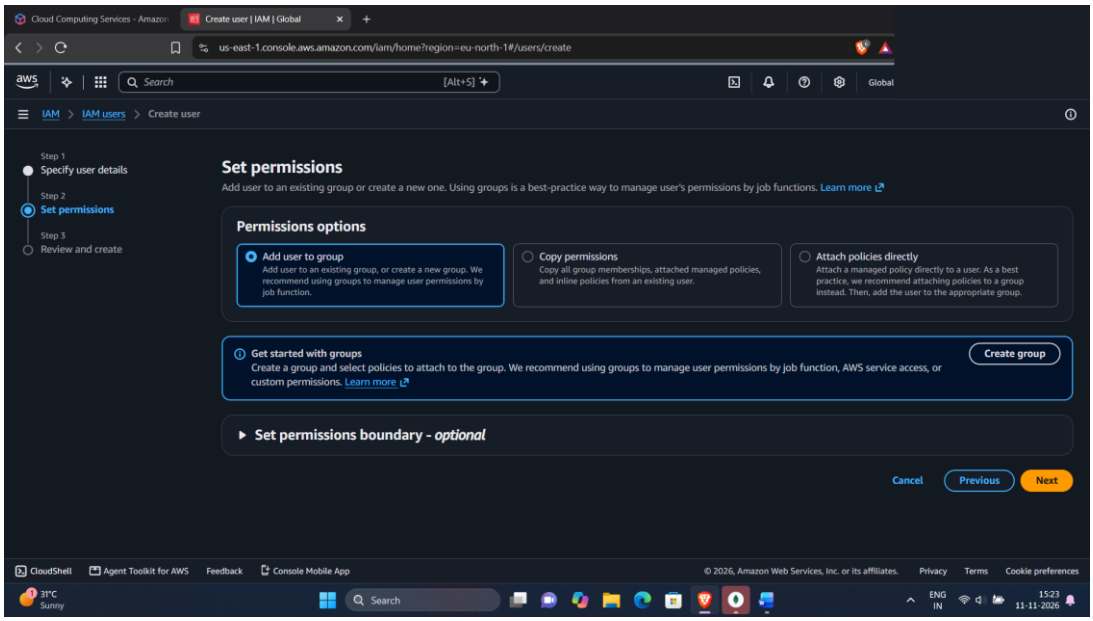


Figure 1. IAM permissions selection during user creation.

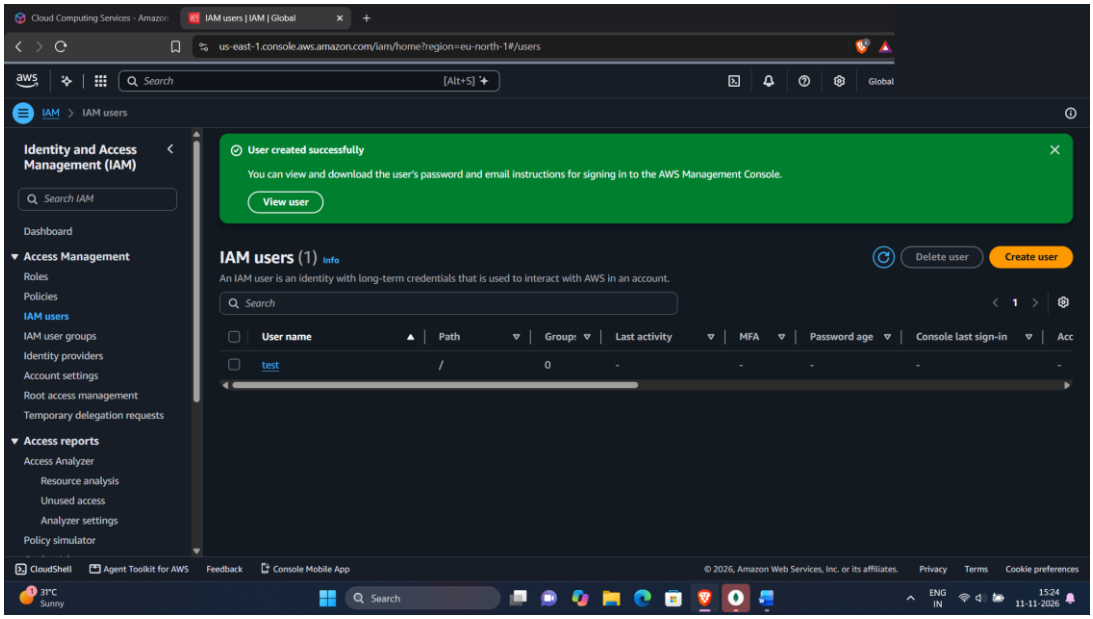


Figure 2. IAM user list showing the created user.

SCREENSHOT EVIDENCE – PROGRAMMATIC ACCESS

The following screenshots document the IAM user details and the selected programmatic-access use case. The user details screen indicates that console access is disabled.

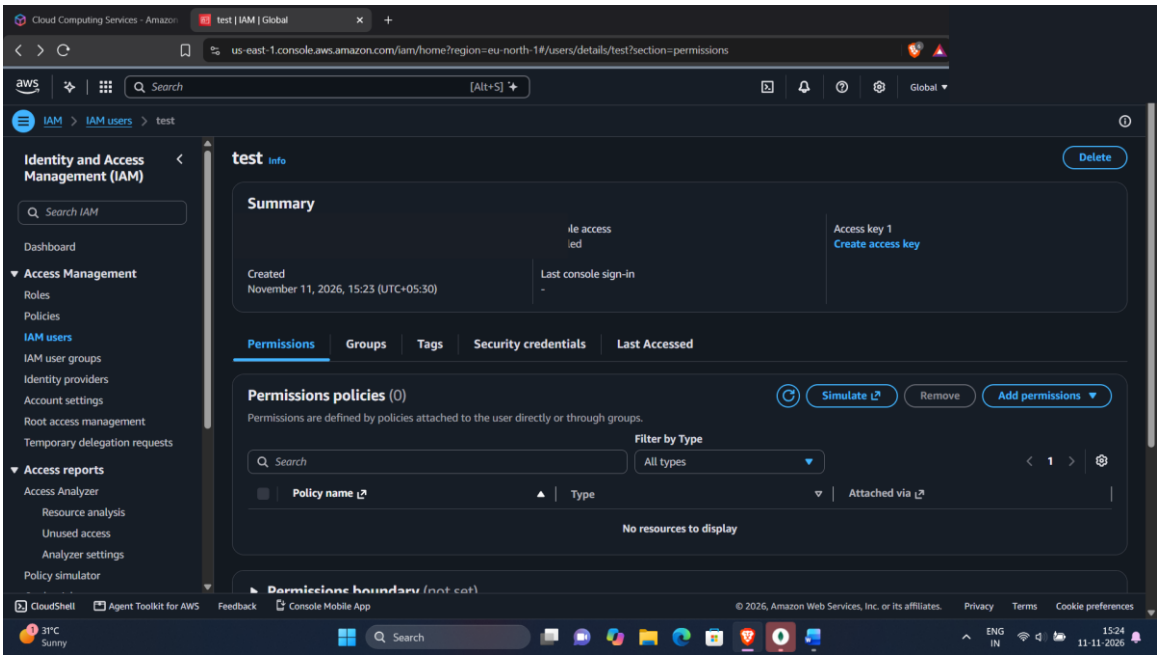


Figure 3. IAM user details; console access is disabled.

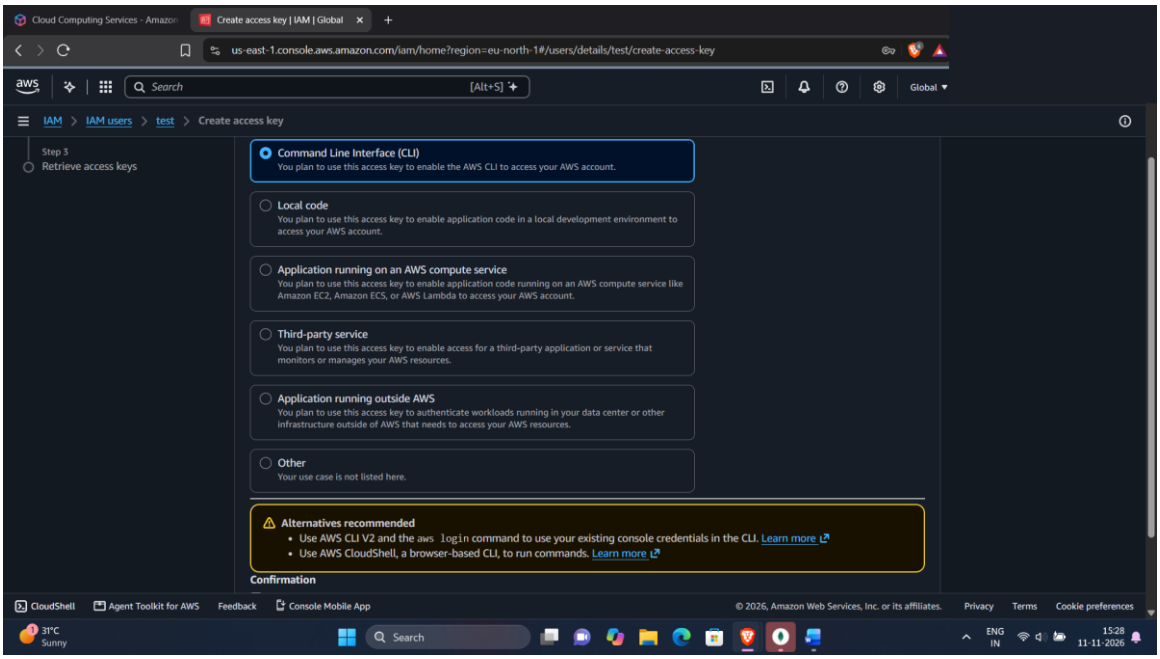


Figure 4. Command Line Interface (CLI) selected as the access-key use case.

ACCESS KEY GENERATION AND SECURE STORAGE

After the key was generated, AWS displayed the access key ID and secret access key and provided the option to download a CSV file. The screenshot below has been redacted to protect the credentials. The actual CSV must be kept privately by the account owner.

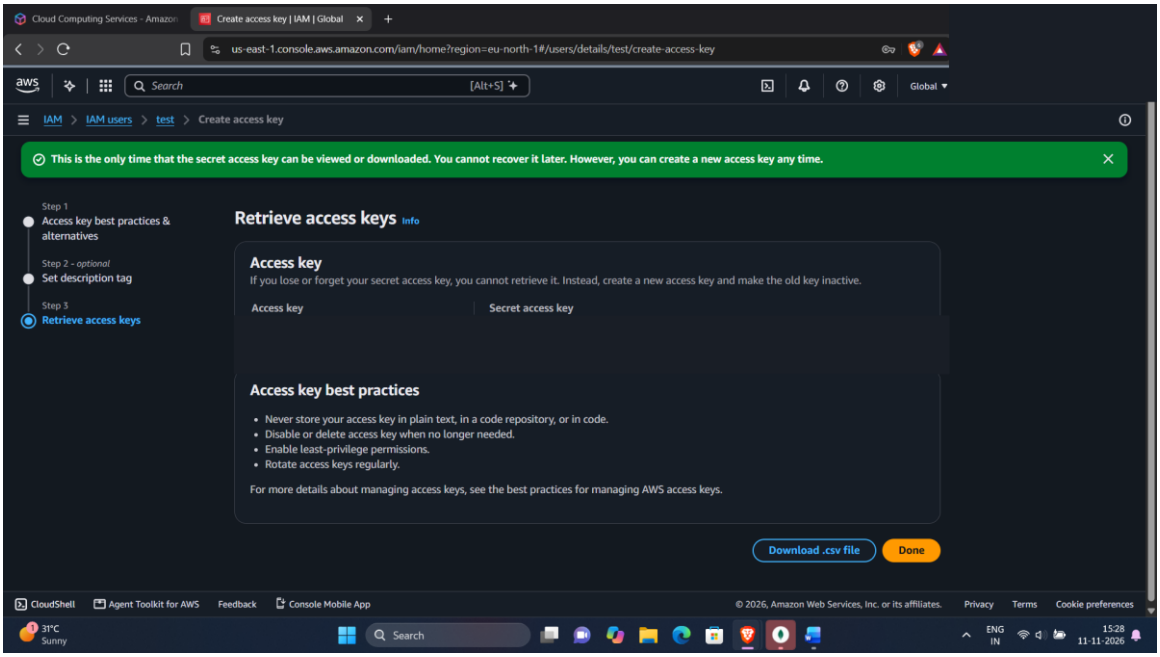


Figure 5. Access-key retrieval page (credential values intentionally redacted).

SECURITY PRECAUTIONS

- Never share the secret access key or upload the unredacted CSV to a report, public drive, Git repository, or chat.
- Use least-privilege permissions and rotate or deactivate keys that are no longer required.
- If the secret key has been exposed, deactivate/delete that key immediately and create a replacement.
- Do not enable console access for a programmatic-only IAM user.

BILLING / FREE TIER ALERT CONFIGURATION

AWS Budgets provides spending notifications. The supplied screenshots show selection of the Zero spend budget template and a budget entry appearing in the Budgets list.

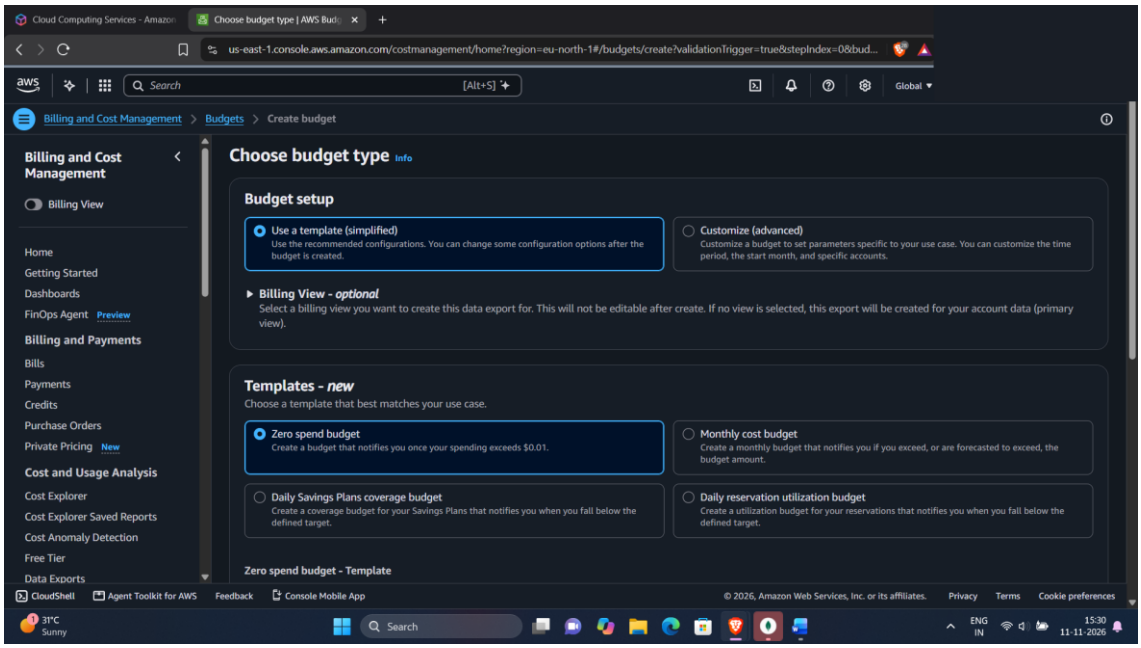


Figure 6. Zero spend budget template selected.

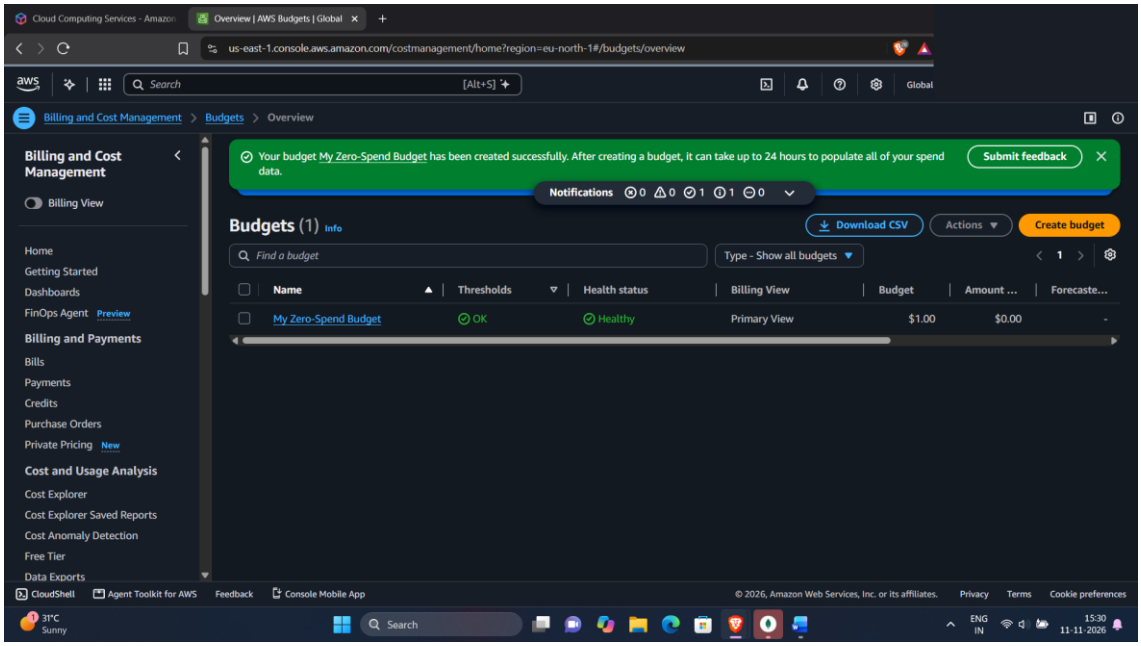


Figure 7. Budget listed in AWS Budgets after creation.

IMPORTANT VERIFICATION NOTE

The supplied screenshots demonstrate that a Zero spend budget was selected and that a budget entry was created. They do not show the notification-recipient field, an enabled Free Tier usage-alert preference, or a delivered test/confirmation email. Therefore, the email-alert portion should be marked fully verified only after the account owner confirms the recipient address, enables Free Tier usage alerts in Billing preferences, and checks the inbox (including Spam/Junk). A Zero spend budget is a spending alert; it is not, by itself, proof that Free Tier usage-threshold alerts are enabled.

OUTPUT / RESULT

An IAM user was created with console access disabled, and the access-key retrieval/download stage is documented with sensitive values redacted. The supplied AWS evidence also shows a Zero spend budget created/listed. The Free Tier email-notification delivery remains to be verified from the account's Billing preferences and notification inbox.

CONCLUSION

The lab demonstrates IAM user creation for programmatic access and the initial setup of an AWS spending budget. Secure handling of access keys and verification of the Free Tier email-alert subscription are essential before considering the entire task complete.